

Scalar analytic branch – current-status and naming alignment

TECT verification-first repository · claim A1-SCALAR-ANALYTIC-BRANCH

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Purpose and scope

The theorem and tier are unchanged. Under $Y > 0$, $Z < 0$, and $m_{\text{sh}}^2 > 0$,

$$K(q) = m_{\text{sh}}^2 + Y(|q|^2 - q_\star^2)^2 \geq m_{\text{sh}}^2 > 0,$$

and every periodic-cell spectral minimum satisfies $\lambda_0(L) \geq m_{\text{sh}}^2$.

The current canonical N-001 convention stores `mu2_shell` as this shell mass. Its original `kinetic_`-coefficients routine reconstructs $r_{\text{zero}} = \text{mu2_shell} + Yq_0^4$. Therefore `mu2_shell` and r_{zero} are distinct; the alias “mu2=r” is forbidden. The obsolete contrary sentence described the failed legacy template, not the current canonical solver manifest.

The current verifier v1.7.0 passes 14/14 overall, including the two named positivity assertions. This card is the scalar A2/A3/A4 positivity input. It does not identify the scalar ‘0.005’ shell mass with the full-production ‘0.2600000000009475’ anchor.

Devil’s-advocate

1. The lower bound fails if the shell mass is nonpositive. The theorem is explicitly conditional on $m_{\text{sh}}^2 > 0$.
2. A finite torus can miss the continuum minimizing shell, but that only raises the spectral minimum; it cannot invalidate $\lambda_0(L) \geq m_{\text{sh}}^2$.
3. The scalar and full-production anchors are numerically different and belong to different theorem branches. This note forbids their identification and makes no full-production measure claim.

Result footer

Result ID:	A1-SCALAR-ANALYTIC-BRANCH
Precise statement:	$Y>0, Z<0, m_{\text{sh}}^2>0$ imply $K \geq m_{\text{sh}}^2$ and $\lambda_{\text{bda0}}(L) \geq m_{\text{sh}}^2$.
Scope:	Positive-mass scalar Brazovskii kernel only.
Dependencies:	A1-KERNEL-IDENTITY; A1-SHELL-POSITIVITY.
Evidence grade:	ANALYTIC, EXECUTED, CONDITIONAL.
Reproduction command:	<code>python codes/foundations/a1_kernel_checks.py</code>
Expected output:	14/14 PASS overall; both positivity assertions PASS.
Falsification gate:	K or λ_{bda0} below the positive shell mass.
Tier before / after:	T6 / T6; administrative alignment only.
No-overclaim statement:	No production config, full functional, dynamics, measure, parameter identity, or BCC conclusion.
Next required action:	Preserve as the scalar A2/A3/A4 positivity input.